Mini Project 2: Tight-Binding Model – Daily Note Draft

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Problem Statement

The electronic structures of two (imaginary) two-dimensional crystals, A and B, are described by a tight-binding model with a two-orbital basis. The dependence of the overlap and hopping parameters, η_{nm} and t_{nm}, on the distance R between lattice sites is given by the following functions:

$$\eta_{11}(R) = \left(1 + \frac{R}{d_0} + \frac{R^2}{3d_0^2}\right) e^{-\frac{R}{d_0}}$$

$$\eta_{12}(R) = \frac{1}{2}\left(1 + \frac{R}{d_0} + \frac{R^2}{64d_0^2}\right) e^{-\frac{R}{2d_0}} - \frac{1}{2} e^{-\frac{R}{d_0}}$$

$$\eta_{22}(R) = \left(1 + \frac{R}{2d_0} + \frac{R^2}{24d_0^2}\right) e^{-\frac{R}{2d_0}}$$

$$t_{11}(R) = -t_0\left(1 + \frac{2R}{d_0}\right) e^{-\frac{R}{d_0}}$$

$$t_{12}(R) = -t_0\left(1 + \frac{2R}{d_0}\right) e^{-\frac{R}{d_0}} + t_0\left(1 + \frac{R}{16d_0}\right) e^{-\frac{R}{2d_0}}$$

$$t_{22}(R) = -t_0\left(1 + \frac{R}{2d_0} - \frac{R^2}{24d_0^2}\right) e^{-\frac{R}{2d_0}}$$

where d_0 and t_0 are positive constants that we use as length and energy units, respectively. The expressions for the hopping parameters are valid for $R > 3d_0$ and we have $t_{nm}(0) = 0$. The atomic energies associated with the two orbitals are $\epsilon_1 = 0$ and $\epsilon_2 = 0.8t_0$.

1. Calculate the electronic band structures of the two crystals assuming that the lattice constant is: (i) $a = 5d_0$; (ii) $a = 7d_0$; (iii) $a = 9d_0$. Include the nearest neighbor and the next nearest neighbor contributions.
2. Compare the results in (1) with the corresponding energy spectra calculated: (i) using only nearest neighbor contributions; (ii) neglecting the overlap integrals, $\eta_{nm}(R) = \delta_{n,m}\delta_{R,0}$.

Note: Specific sets of two-dimensional lattices will be assigned by a random draw at the end of the class on 10/08/2026. Please make sure that the (two-student) project teams are established and that each team has (at least) one representative present for the draw.

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1 General Considerations for the Solution

To compute the band structure, we need to solve the Schrödinger equation in terms of the overlap integrals:

$$[\bar{\epsilon}\bar{\eta}(\vec{k}) + \bar{t}(\vec{k})]\bar{c} = E(\vec{k})\bar{\eta}(\vec{k})\bar{c}$$

where the Hermitian matrices $\bar{\eta}$ and $\bar{t}$ are matrices with the following matrix elements:

$$\eta_{nm}(\vec{k}) = \sum_{\vec{R}_j} \eta_{nm}(|\vec{R}_j + (\vec{\tau}_m - \vec{\tau}_n)|)e^{i\vec{k}\cdot\vec{R}_j} \tag{1}$$

$$t_{nm}(\vec{k}) = \sum_{\vec{R}_j} t_{nm}(|\vec{R}_j + (\vec{\tau}_m - \vec{\tau}_n)|)e^{i\vec{k}\cdot\vec{R}_j} \tag{2}$$

where $\vec{R}_j$ is a vector on the Bravais lattice, and $\vec{\tau}_n, \vec{\tau}_m$ are the sub-lattice basis vectors. $\bar{\epsilon}$ is the diagonal matrix of on-site energies for each orbital.

We can define $\bar{c}' = \bar{\eta}(\vec{k})\bar{c}$ (such that $\bar{c} = \bar{\eta}^{-1}(\vec{k})\bar{c}'$) and transform the equation into a standard eigenvalue problem:

$$[\bar{\epsilon} + \bar{t}(\vec{k})\bar{\eta}^{-1}(\vec{k})]\bar{c}' = E(\vec{k})\bar{c}'$$

The procedure for both crystal A and crystal B is to determine the relevant $\bar{\epsilon}$, $\bar{\eta}(\vec{k})$, and $\bar{t}(\vec{k})$, compute the Fourier transformed expressions, and then solve the eigenvalue problem as a function of $\vec{k}$ to extract the band structure.

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2 Crystal A: Square Lattice

The square lattice has no lattice basis, therefore the only relevant $\vec{\tau}$ is $\vec{\tau} = (0, 0)$, and there are only 2 orbitals. Therefore, the eigenvalue problem as a function of $\vec{k}$ is 2×2, corresponding to the two orbitals on-site.

Therefore, $\bar{\epsilon}$ is given by:

$$\bar{\epsilon} = \begin{pmatrix} 0 & 0 \\ 0 & 0.8t_0 \end{pmatrix}$$

with $\bar{t}$ and $\bar{\eta}$ defined as 2×2 Hermitian matrices:

$$\bar{t}(\vec{k}) = \begin{pmatrix} t_{11}(\vec{k}) & t_{12}(\vec{k}) \\ t_{12}^*(\vec{k}) & t_{22}(\vec{k}) \end{pmatrix}, \qquad \bar{\eta}(\vec{k}) = \begin{pmatrix} \eta_{11}(\vec{k}) & \eta_{12}(\vec{k}) \\ \eta_{12}^*(\vec{k}) & \eta_{22}(\vec{k}) \end{pmatrix}$$

On-Site Contribution ($R = 0$)

The on-site contribution gives the identity matrix:

$$\eta_{nm}^{(0)}(\vec{k}) = \eta_{nm}(0) = \delta_{nm}, \qquad t_{nm}^{(0)}(\vec{k}) = 0$$

Nearest Neighbor Contributions

For the square lattice, this is straightforward. We consider one orbital pair as an example, and the rest follows by the same logic.

Using the set of nearest-neighbor vectors $\vec{R}_j = (\pm a, 0), (0, \pm a)$ with distance $R = a$, since η_{nm} and t_{nm} depend only on distance R:

$$\begin{aligned}\eta_{nm}^{(1)}(\vec{k}) &= \sum_{\vec{R}_{\mathrm{NN}}} \eta_{nm}(\vec{R}_j) e^{i\vec{k}\cdot\vec{R}_j} = \eta_{nm}(a)\left[e^{ik_x a} + e^{-ik_x a} + e^{ik_y a} + e^{-ik_y a}\right] \\ &= 2\eta_{nm}(a)\left[\cos(k_x a) + \cos(k_y a)\right] = 2\eta_{nm}(a)\epsilon_0(\vec{k})\end{aligned}$$

where $\epsilon_0(\vec{k}) = \cos(k_x a) + \cos(k_y a)$. This factor is the same across all orbital pairs, so in general we can write:

$$\begin{aligned}\eta_{nm}^{(1)}(\vec{k}) &= 2\eta_{nm}(a)\epsilon_0(\vec{k}) \\ t_{nm}^{(1)}(\vec{k}) &= 2t_{nm}(a)\epsilon_0(\vec{k})\end{aligned}$$

Next-Nearest Neighbor Contributions

A similar logic follows for the next-nearest neighbors at $\vec{R}_j = (\pm a, \pm a)$ with distance $R = \sqrt{2}a$:

$$\begin{aligned}\eta_{nm}^{(2)}(\vec{k}) &= \eta_{nm}(\sqrt{2}a)\left[e^{ia(k_x+k_y)} + e^{ia(k_x-k_y)} + e^{-ia(k_x-k_y)} + e^{-ia(k_x+k_y)}\right] \\ &= 2\eta_{nm}(\sqrt{2}a)\left[\cos(ak_x + ak_y) + \cos(ak_x - ak_y)\right] \\ &= 4\eta_{nm}(\sqrt{2}a)\cos(k_x a)\cos(k_y a) = 2\eta_{nm}(\sqrt{2}a)\epsilon_1(\vec{k})\end{aligned}$$

where $\epsilon_1(\vec{k}) = \cos(a(k_x + k_y)) + \cos(a(k_x - k_y)) = 2\cos(k_x a)\cos(k_y a)$.

The exact same is true for t_{nm}, so including the on-site term δ_{nm} for the overlap, we can write the total contributions as:

$$\begin{aligned}\eta_{nm}(\vec{k}) &= \delta_{nm} + 2\eta_{nm}(a)\epsilon_0(\vec{k}) + 2\eta_{nm}(\sqrt{2}a)\epsilon_1(\vec{k}) \\ t_{nm}(\vec{k}) &= 2t_{nm}(a)\epsilon_0(\vec{k}) + 2t_{nm}(\sqrt{2}a)\epsilon_1(\vec{k})\end{aligned}$$

$$\begin{aligned}\epsilon_0(\vec{k}) &= \cos(k_x a) + \cos(k_y a) \\ \epsilon_1(\vec{k}) &= 2\cos(k_x a)\cos(k_y a)\end{aligned}$$

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3 Crystal B: Centered Rectangular Lattice

Crystal B is modeled on a centered rectangular lattice with a two-site basis:

- Site 1 at $\vec{\tau}_1 = (0, 0)$, which has two orbitals (orbital 1 and orbital 2)
- Site 2 at $\vec{\tau}_2 = \left(\frac{a}{2}, \frac{b}{2}\right)$, which has one orbital (orbital 1)

This gives a 3-orbital basis:

1. Orbital 1 on Site 1 ($\epsilon_1 = 0$)
2. Orbital 2 on Site 1 ($\epsilon_2 = 0.8t_0$)
3. Orbital 1 on Site 2 ($\epsilon_3 = \epsilon_1 = 0$)

Since State 1 is orbital 1 (Site 1), State 2 is orbital 2 (Site 1), and State 3 is orbital 1 (Site 2), the off-diagonal inter-sublattice couplings map directly to the radial functions:

$$\eta_{13}(R) \equiv \eta_{11}(R), \qquad t_{13}(R) \equiv t_{11}(R)$$
$$\eta_{23}(R) \equiv \eta_{12}(R), \qquad t_{23}(R) \equiv t_{12}(R)$$

Therefore, the eigenvalue problem as a function of $\vec{k}$ is 3×3, and $\bar{\epsilon}$ is given by:

$$\bar{\epsilon} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0.8t_0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

with $\bar{t}$ and $\bar{\eta}$ defined as 3×3 Hermitian matrices:

$$\bar{t}(\vec{k}) = \begin{pmatrix} t_{11}(\vec{k}) & t_{12}(\vec{k}) & t_{13}(\vec{k}) \\ t_{12}^*(\vec{k}) & t_{22}(\vec{k}) & t_{23}(\vec{k}) \\ t_{13}^*(\vec{k}) & t_{23}^*(\vec{k}) & t_{33}(\vec{k}) \end{pmatrix}, \qquad \bar{\eta}(\vec{k}) = \begin{pmatrix} \eta_{11}(\vec{k}) & \eta_{12}(\vec{k}) & \eta_{13}(\vec{k}) \\ \eta_{12}^*(\vec{k}) & \eta_{22}(\vec{k}) & \eta_{23}(\vec{k}) \\ \eta_{13}^*(\vec{k}) & \eta_{23}^*(\vec{k}) & \eta_{33}(\vec{k}) \end{pmatrix}$$

For this lattice, contributions are classified by the actual physical interatomic distance $R = |\vec{R}_j + (\vec{\tau}_m - \vec{\tau}_n)|$ into on-site, nearest-neighbor, and next-nearest-neighbor shells.

1. On-Site ($R = 0$) Contribution

Because Site 1 at $\vec{\tau}_1 = (0, 0)$ and Site 2 at $\vec{\tau}_2 = \left(\frac{a}{2}, \frac{b}{2}\right)$ are physically distinct atoms, there is no $R = 0$ inter-sublattice term ($\eta_{13}^{(0)} = \eta_{23}^{(0)} = 0$ and $t_{nm}^{(0)} = 0$). The on-site overlap matrix is purely diagonal:

$$\eta_{nm}^{(0)}(\vec{k}) = \eta_{nm}(0) = \delta_{nm}, \qquad t_{nm}^{(0)}(\vec{k}) = 0$$

Specifically:

$$\eta_{11}^{(0)} = \eta_{22}^{(0)} = \eta_{33}^{(0)} = 1, \qquad \eta_{12}^{(0)} = \eta_{13}^{(0)} = \eta_{23}^{(0)} = 0$$

2. Nearest Neighbor Contributions '(1)'

Intra-Sublattice Elements ($11, 22, 33$ and 12)

For the intra-sublattice terms ($\vec{\tau}_m - \vec{\tau}_n = \vec{0}$), the nearest neighbors on the Bravais lattice lie along the axes at $\vec{R}_j = (\pm a, 0)$ and $(0, \pm b)$ with distances $R = a$ and $R = b$, respectively.

For the diagonal elements:

$$\eta_{nn}^{(1)}(\vec{k}) = 2\eta_{nn}(a)\cos(k_x a) + 2\eta_{nn}(b)\cos(k_y b)$$
$$t_{nn}^{(1)}(\vec{k}) = 2t_{nn}(a)\cos(k_x a) + 2t_{nn}(b)\cos(k_y b)$$

where for State 3 (orbital 1 on Site 2), $\eta_{33}^{(1)} = \eta_{11}^{(1)}$ and $t_{33}^{(1)} = t_{11}^{(1)}$.

For the off-diagonal intra-sublattice elements (η_{12}, t_{12}), since orbitals 1 and 2 reside on the same site, the on-site overlap is zero by orthogonality. The nearest-neighbor interaction yields:

$$\eta_{12}^{(1)}(\vec{k}) = 2\eta_{12}(a)\cos(k_x a) + 2\eta_{12}(b)\cos(k_y b)$$
$$t_{12}^{(1)}(\vec{k}) = 2t_{12}(a)\cos(k_x a) + 2t_{12}(b)\cos(k_y b)$$

Inter-Sublattice Elements (η_{13}, t_{13} and η_{23}, t_{23})

For the inter-sublattice couplings, $\vec{\tau}_2 - \vec{\tau}_1 = \left(\frac{a}{2}, \frac{b}{2}\right)$. The four nearest-neighbor Site 2 atoms surrounding Site 1 lie symmetrically at displacements $\vec{d} = \left(\pm\frac{a}{2}, \pm\frac{b}{2}\right)$, all at the exact same physical distance $R = \frac{1}{2}\sqrt{a^2 + b^2}$. These correspond to the four Bravais lattice vectors:

$$\vec{R}_j \in \{(0,0), (-a,0), (0,-b), (-a,-b)\}$$

$$\begin{aligned}\eta_{13}^{(1)}(\vec{k}) &= \eta_{13}\left(\frac{\sqrt{a^2+b^2}}{2}\right)\left[1 + e^{-iak_x} + e^{-ibk_y} + e^{-i(ak_x+bk_y)}\right] \\ &= \eta_{13}\left(\frac{\sqrt{a^2+b^2}}{2}\right)\left(1 + e^{-iak_x}\right)\left(1 + e^{-ibk_y}\right)\end{aligned}$$

$$t_{13}^{(1)}(\vec{k}) = t_{13}\left(\frac{\sqrt{a^2+b^2}}{2}\right)\left(1 + e^{-iak_x}\right)\left(1 + e^{-ibk_y}\right)$$

$$\eta_{23}^{(1)}(\vec{k}) = \eta_{23}\left(\frac{\sqrt{a^2+b^2}}{2}\right)\left(1 + e^{-iak_x}\right)\left(1 + e^{-ibk_y}\right)$$

$$t_{23}^{(1)}(\vec{k}) = t_{23}\left(\frac{\sqrt{a^2+b^2}}{2}\right)\left(1 + e^{-iak_x}\right)\left(1 + e^{-ibk_y}\right)$$

3. Next-Nearest Neighbor Contributions '(2)'

Intra-Sublattice Elements ($11, 22, 33$ and 12)

Displacements to the four corner unit cells lie at $\vec{R}_j = (\pm a, \pm b)$ with distance $R = \sqrt{a^2 + b^2}$:

$$\eta_{nn}^{(2)}(\vec{k}) = 4\eta_{nn}\left(\sqrt{a^2+b^2}\right)\cos(ak_x)\cos(bk_y)$$
$$t_{nn}^{(2)}(\vec{k}) = 4t_{nn}\left(\sqrt{a^2+b^2}\right)\cos(ak_x)\cos(bk_y)$$

$$\eta_{12}^{(2)}(\vec{k}) = 4\eta_{12}\left(\sqrt{a^2+b^2}\right)\cos(ak_x)\cos(bk_y)$$
$$t_{12}^{(2)}(\vec{k}) = 4t_{12}\left(\sqrt{a^2+b^2}\right)\cos(ak_x)\cos(bk_y)$$

Inter-Sublattice Elements (η_{13}, t_{13} and η_{23}, t_{23})

The second inter-sublattice shell consists of 8 neighbors formed by:

- 4 neighbors extended along x at displacements $\vec{d} = \left(\pm\frac{3a}{2}, \pm\frac{b}{2}\right)$ with distance $R = \frac{1}{2}\sqrt{9a^2+b^2}$, generated by $\vec{R}_j \in \{(a,0), (a,-b), (-2a,0), (-2a,-b)\}$.
- 4 neighbors extended along y at displacements $\vec{d} = \left(\pm\frac{a}{2}, \pm\frac{3b}{2}\right)$ with distance $R = \frac{1}{2}\sqrt{a^2+9b^2}$, generated by $\vec{R}_j \in \{(0,b), (-a,b), (0,-2b), (-a,-2b)\}$.

$$\eta_{13}^{(2)}(\vec{k}) = \eta_{13}\left(\frac{\sqrt{9a^2+b^2}}{2}\right)\left[e^{iak_x} + e^{i(ak_x-bk_y)} + e^{-2iak_x} + e^{-i(2ak_x+bk_y)}\right]$$
$$+ \eta_{13}\left(\frac{\sqrt{a^2+9b^2}}{2}\right)\left[e^{ibk_y} + e^{i(-ak_x+bk_y)} + e^{-2ibk_y} + e^{-i(ak_x+2bk_y)}\right]$$

$$t_{13}^{(2)}(\vec{k}) = t_{13}\left(\frac{\sqrt{9a^2+b^2}}{2}\right)\left[e^{iak_x} + e^{i(ak_x-bk_y)} + e^{-2iak_x} + e^{-i(2ak_x+bk_y)}\right]$$
$$+ t_{13}\left(\frac{\sqrt{a^2+9b^2}}{2}\right)\left[e^{ibk_y} + e^{i(-ak_x+bk_y)} + e^{-2ibk_y} + e^{-i(ak_x+2bk_y)}\right]$$

$$\eta_{23}^{(2)}(\vec{k}) = \eta_{23}\left(\frac{\sqrt{9a^2+b^2}}{2}\right)\left[e^{iak_x} + e^{i(ak_x-bk_y)} + e^{-2iak_x} + e^{-i(2ak_x+bk_y)}\right]$$
$$+ \eta_{23}\left(\frac{\sqrt{a^2+9b^2}}{2}\right)\left[e^{ibk_y} + e^{i(-ak_x+bk_y)} + e^{-2ibk_y} + e^{-i(ak_x+2bk_y)}\right]$$

$$t_{23}^{(2)}(\vec{k}) = t_{23}\left(\frac{\sqrt{9a^2+b^2}}{2}\right)\left[e^{iak_x} + e^{i(ak_x-bk_y)} + e^{-2iak_x} + e^{-i(2ak_x+bk_y)}\right]$$
$$+ t_{23}\left(\frac{\sqrt{a^2+9b^2}}{2}\right)\left[e^{ibk_y} + e^{i(-ak_x+bk_y)} + e^{-2ibk_y} + e^{-i(ak_x+2bk_y)}\right]$$

The lower-triangle elements are given by the Hermitian conjugate relations:

$$t_{ji}(\vec{k}) = t_{ij}^*(\vec{k}), \qquad \eta_{ji}(\vec{k}) = \eta_{ij}^*(\vec{k})$$

4. Total Matrix Assembly & Approximations

The total matrix elements are combined as:

$$\eta_{nm}(\vec{k}) = \eta_{nm}^{(0)}(\vec{k}) + \eta_{nm}^{(1)}(\vec{k}) + \eta_{nm}^{(2)}(\vec{k})$$
$$t_{nm}(\vec{k}) = t_{nm}^{(0)}(\vec{k}) + t_{nm}^{(1)}(\vec{k}) + t_{nm}^{(2)}(\vec{k})$$